

14. Let R_1 and R_2 be relations on a set A represented by the matrices

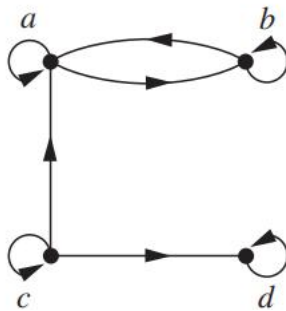
$$\mathbf{M}_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Find the matrices that represent

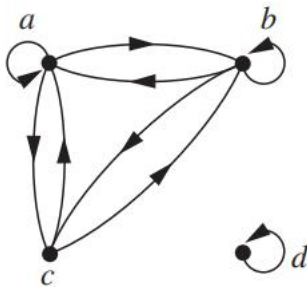
- a) $R_1 \cup R_2$. b) $R_1 \cap R_2$. c) $R_2 \circ R_1$.
d) $R_1 \circ R_1$. e) $R_1 \oplus R_2$.

Determine whether the relations represented by the directed graphs shown in Exercises 26–28 are reflexive, irreflexive, symmetric, antisymmetric, asymmetric, and/or transitive.

26.



27.



28.

